

Qatar-10b

R-1910

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_260701 · created 2026-08-05T23:35:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461222.651 +/- 0.015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.031 +/- 0.049
Transit depth (Rp/Rs)^2	0.094 +/- 0.3 [%]
Orbital Inclination (inc)	86.26 +/- 2.66
Airmass coefficient 1 (a1)	1.314248 +/- 3.2e-05
Airmass coefficient 2 (a2)	-0.204 +/- 0.066
Scatter in the residuals of the lightcurve fit is	0.00067 %
Best Comparison Star	#1 - [236, 299]
Optimal Aperture	2.88
Optimal Annulus	3.75
Transit Duration (day)	0.0027 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.031 ± 0.049 vs 0.1265 (deviation 1.95σ)
Duration fit vs published	0.0027 d vs 0.1154 d (deviation 18.48σ)
Mid-time O–C	-9.7 min
Depth SNR	0.6
Residual scatter	0.00067 %

Light curve

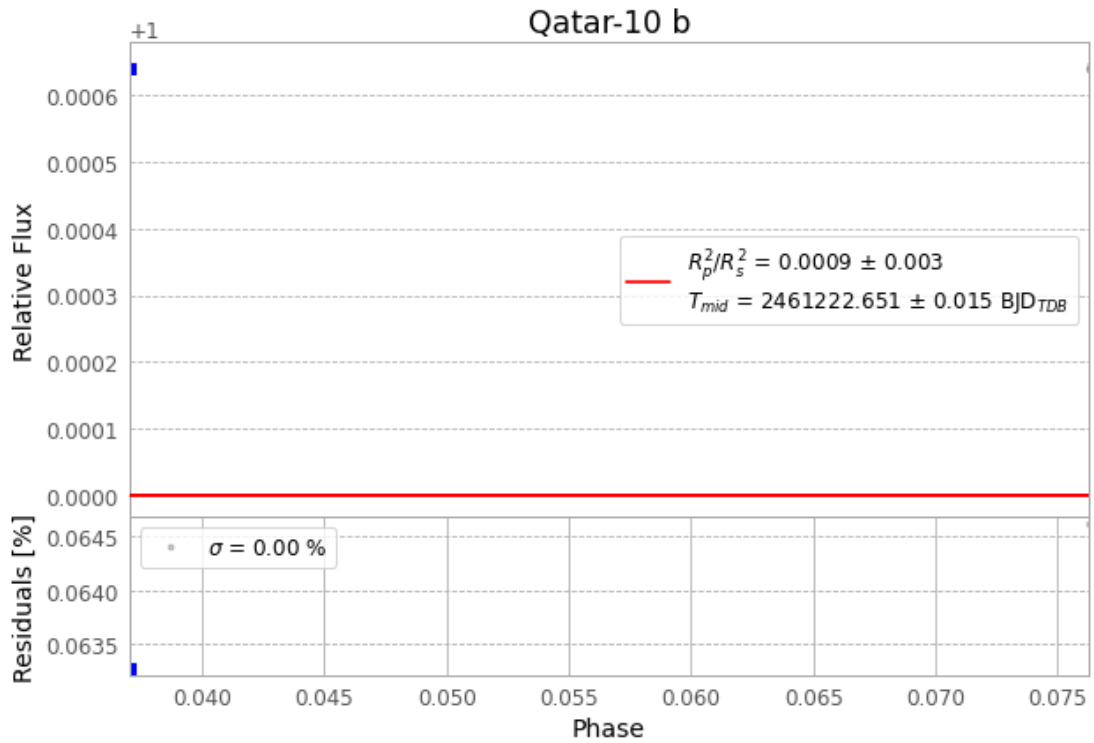


Figure 1 — FinalLightCurve_Qatar-10 b_2026-06-30.png (SHA-256 f8e8b61c1de78199...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [304, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5a9fe0fc2a5cfa66...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

137 FITS frames (56 MB), Qatar-10260701030117.FITS.gz → Qatar-10260701065115.FITS.gz. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-260701.log (36158b56fa9b...), FinalLightCurve_Qatar-10 b_2026-06-30.png (f8e8b61c1de7...), FinalLightCurve_Qatar-10 b_2026-06-30.csv (bf06d4026e58...)

Compute resources

Wall time 203.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:35Z.